

UNIT II MACRONUTRIENTS

- Classification, functions, Digestion & Absorption-
 - *Carbohydrates
 - *Protein
 - *Fat
- RDA (Recommended Dietary Allowances),
- Sources of Protein Carbohydrate and Fat,
- Deficiencies of Protein Carbohydrate and Fat,
- Dietary fiber - types and functions

- Nutrients are organic molecules that are necessary for all organisms to grow, make energy, develop, and reproduce.
- They are broadly classified into two; macronutrients and micronutrients.
- Macronutrients: these are required in large amounts and are further classified into three types; carbohydrates, proteins, and fats.
- • Micronutrients: these are required in small quantities and they consist of vitamins and minerals.
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Nutrients

Macronutrients

Required in relatively larger amounts



Carbohydrates

Provide energy; support digestive health and immune function



Lipids (fats)

Support cellular function and structure; regulate temperature; protect body organs; store energy in the body



Protein

Regulate cellular processes; support mechanical and structural function; provide energy



Vitamins

Support cell function, development, and growth; function as antioxidants; assist the absorption of



Minerals

Support bone, muscular, cardiovascular, and nervous system functions; produce enzymes and hormones

Micronutrients

Required in relatively smaller amounts



Phytochemicals/ phytonutrients

(Not considered essential)

May help prevent chronic diseases; exert actions such as anti-inflammatory, anti-microbial, and antioxidant effects

Macronutrients - Macronutrients are nutrients needed in large amounts for energy, growth, and bodily functions, primarily classified into three main types

- Carbohydrates (energy source, ~ 4 kcal/ g),
 - Proteins (building blocks, ~ 4 kcal/ g), and
 - Fats/ Lipids (concentrated energy, ~ 9 kcal/ g).
- While water is also needed in large amounts, it's not a calorie provider but crucial for bodily processes

- Carbohydrates

- Carbohydrates are molecules composed of carbon, hydrogen, and oxygen.
- Carbon + water molecules = Hydrated carbon = Carbo-hydrate
- The major food sources of carbohydrates are grains, dairy products, fruits, legumes, and starchy vegetables, like potatoes.

- Carbohydrates are a major fuel source for all cells of the body, and certain cells, like cells of the central nervous system and red blood cells, rely solely on carbohydrates for energy.
- Carbohydrates are the chief sources of energy (Providing 4 Kcal/gm)

- Carbohydrates are broadly classified into two groups based on their chemical structure:
 - simple carbohydrates (often called simple sugars) and
 - complex carbohydrates, which include fiber, starch, and glycogen.
- Carbohydrates are divided into three major groups:
 - Sugars
 - Starches
 - Cellulose and related material (fiber).

Sugar

- Sugar is the simplest type of carbohydrate that provides energy rapidly.
- Our body quickly breaks down into glucose and absorbs simple carbohydrates, which results in the fluctuation of blood sugar levels.
- Glucose is the crucial energy source for our body's cells, tissues, and organs.
- Examples are white sugar, honey, glucose, cane syrup etc.

There are two types of sugars

- Naturally occurring sugars Milk and fresh fruits
- Added sugars Sweets (cookies, candy bars, ice cream), canned fruit, juice and soda

Properties of sugars

- All sugars (monosaccharides and disaccharides) dissolve in water and are varying sweet in taste.
- They form white or colourless crystals and their taste can be modified by cooking (caramelization).
- Sugars are a readily available source of energy and sweetness and also widely used in food industry like jam making, canning, freezing (as preservatives), biscuits, cakes, soft drinks, chocolates (to provide texture and consistency).

Classification of sugar:

- Monosaccharides: these are the simplest form of carbohydrates containing simple sugar molecule. Example: Glucose, Fructose and Galactose.
- Disaccharides: These carbohydrates composed of two units of Monosaccharides. Example: Sucrose, Lactose and Maltose.

Monosaccharide or simple sugar:

- Glucose, also known as dextrose is found naturally in fruits and plants juices and in the blood of living animals
- Most carbohydrates in food are ultimately converted to glucose after digestion. Glucose can also be
- manufactured from starch by the action of acid or specific enzymes. Artificially manufactured glucose syrups also known as liquid glucose are used to manufacture foods like soft drinks, jams and sugar confectionary.
- Fructose also occurs naturally in some fruits and vegetables and especially in honey. It is the sweetest sugar. It is also a component of sucrose and is also present in commercial 'high fructose' syrups.
- Galactose is a part of lactose and does not occur in free state.

Disaccharides

- Disaccharides consists of two monosaccharides linked together minus the elements of water .
- Sucrose occurs naturally in sugar cane and sugar beet and in lesser amounts in fruits and some root vegetables like carrots. It is chemical combination of glucose and fructose. The 'sugar' that we eat, whether white or brown and essentially pure sucrose.
- Maltose is a combination of 2 glucose units and is formed during the breakdown of starch by digestion.
- Lactose is a combination of glucose and galactose and occurs only in milk, including human milk. It is less sweet than sucrose or glucose.

Star ch

- Star ch (amylum), a natural component of plants, is a complex carbohydrate.
- Starches are polysaccharides are composed of large number of glucose units linked together to form both straight and branched chains.
- It is a natural polymer consisting of a long chain of glucose molecules. The body breaks down the natural polymer into glucose- the energy source for the human body.
- Blood sugar remains stable because the body takes a long time to break down this complex carbohydrate.

- There are two types: amylose and amylopectin.
 - Amylose is a linear polymer, which is amorphous or solid.
- Amylopectin is a branched chain polymer, which is crystalline.
- It is present in many foods, such as grains, vegetables, and fruits (apples, berries, melons).
- Examples are maize, tapioca, brown rice, oatmeal, roots, tubers, whole-wheat bread, pasta, corn, peas, wheat, potatoes, legumes, beans (black beans, chickpeas, lentils, lima, beans, and kidney beans), etc.

Fiber

- Fiber, a complex healthy carbohydrate or non-starch polysaccharide, does not break down glucose.
- Fiber can pass through the intestines and stimulate digestion. It is present in plant-based foods such as vegetables, fruits, and whole grains.
- According to the FDA (Food and Drug Administration), females should consume 28 grams (g) per day, whereas males should consume 34 grams.

- They are of two types: soluble and insoluble.
- Soluble fibers can dissolve in water, whereas insoluble one cannot.
- It helps to regulate blood sugar levels, feeds good bacteria in the gut, helps slow digestion, softens stool, and lowers cholesterol.
- Examples of high-fiber foods are Beans and legumes (black beans, chickpeas, lentils, lima beans, peanuts, and pinto beans), fruits (apples, seeds, and peaches), nuts and seeds (almonds, walnuts, pumpkin seeds, and sunflower seeds), whole-grain products (almonds, walnuts, pumpkin seeds, and sunflower seeds), and vegetables (corn, broccoli, Brussels sprouts and squash).

Daily requirements

- Carbohydrate intake should be in the range of 300-500gm (50%- 70%) out of the total energy intake for adults and 40-60% for children.

Sources

- All sugars, jaggery, honey, pulses, whole grains, cereals, grains, rice, fruits, milk, yogurt, beans, roots and tubers such as potatoes, beet root etc.
- Sugars in diet are sugar, sweets, chocolates, milk, fruit, fruit products, biscuits, cakes
- Starches are bread, potatoes, cereal products
- Fibre are vegetables and cereal products

Digestion and absorption of Carbohydrates

- Salivary amylase aids digestion of starch in the mouth.
- Most of the digestion of carbohydrates takes place in the small intestine.
- Dietary carbohydrates, which are primarily present as polysaccharides (starch) followed by disaccharide and free monosaccharide, supply 60-70% of energy to the body.
- To be absorbed from the gut, carbohydrates must be broken down to their constituent monosaccharide units

Let's see how these carbohydrates are digested in the gut –

- Glucose and galactose are absorbed into the mucosal cells. Glucose and galactose cannot attach to the carrier until it has been preloaded with sodium.
- Hence, we would realize that oral rehydration syrup (ORS) always contains sodium chloride and glucose / sugar.
- Fructose is absorbed by a facilitated transport, involving a specific transporter - GLUT.
- GLUTS or human glucose transporters are the integral proteins, which penetrate and span the lipid bilayer of plasma membrane.

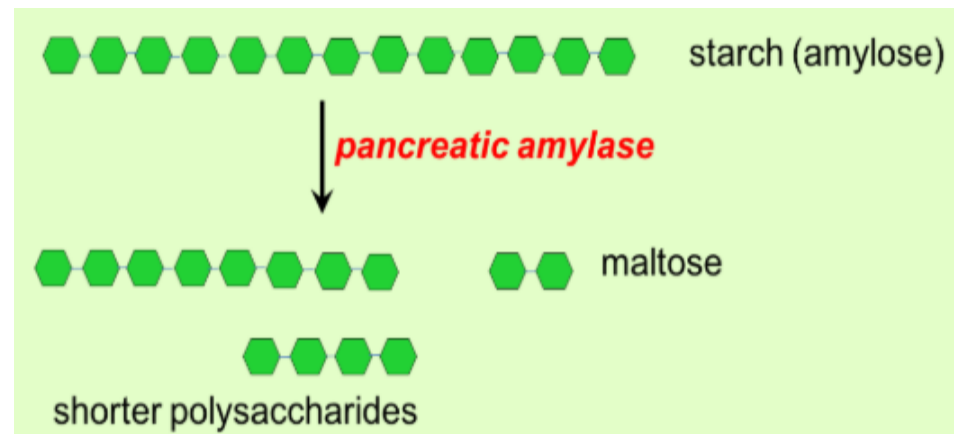
- Following absorption, the monosaccharides enter the portal circulation and are carried to the liver.
- Both galactose and fructose are converted to glucose in the hepatocytes
- A part of glucose is converted into glycogen while some is catabolized for energy in the liver.

- The remainder of glucose passes on into the systemic blood supply and is distributed among other tissues such as skeletal muscle, adipose tissue and kidney.
- Carbohydrates are absorbed into the blood stream as glucose, Galactose and fructose.
- By way of the capillaries of the villi, the simple sugars enter the portal circulation.

- Mouth or Oral cavity (Salivary Amylase)

- Stomach-The low pH in the stomach inactivates salivary amylase

- Small Intestine



[1] Mouth or Oral Cavity –

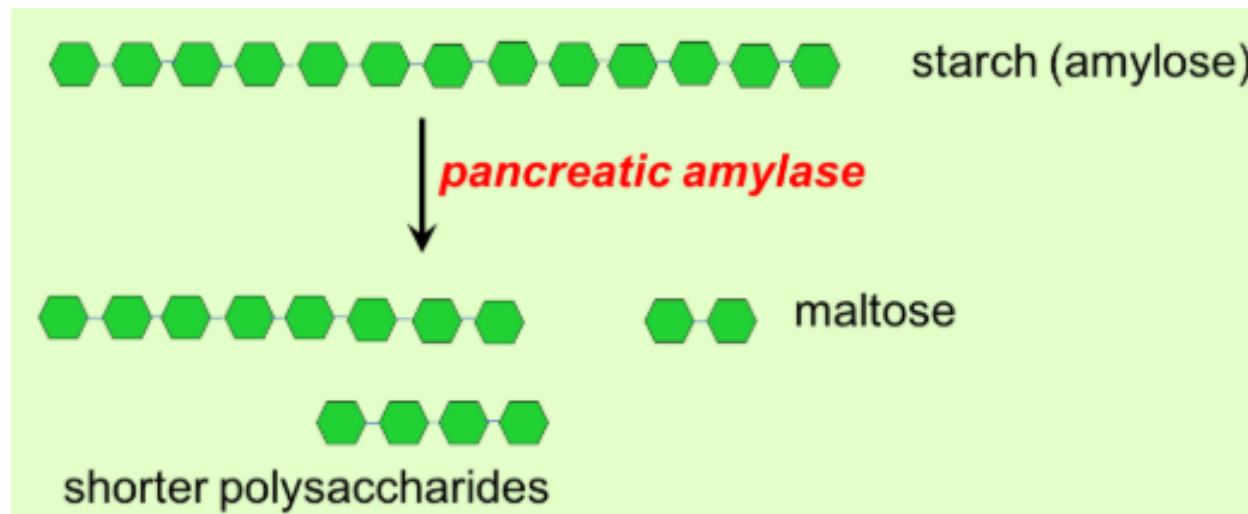
- Mechanical digestion begins that breaks food containing carbohydrate into smaller pieces and mix it with saliva, produced by several salivary glands in the oral cavity.
- Some enzymatic digestion of starch occurs in the mouth, due to the action of the enzyme salivary amylase.
- This enzyme starts to break the long glucose chains of starch into shorter chains, some as small as maltose.

[2] Stomach –

- The low pH in the stomach inactivates salivary amylase, so it no longer works once it arrives at the stomach.
- Although there's more mechanical digestion in the stomach, there's little chemical digestion of carbohydrates here.

[3] Small Intestine –

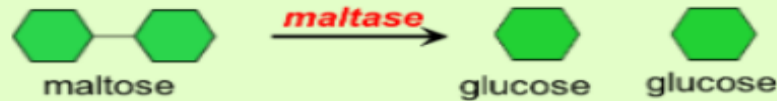
- Most carbohydrate digestion occurs in the small intestine
- Pancreatic amylase is secreted from the pancreas into the small intestine, and like salivary amylase, it breaks starch down to small saccharides (containing 3 to 10 glucose molecules) and maltose.



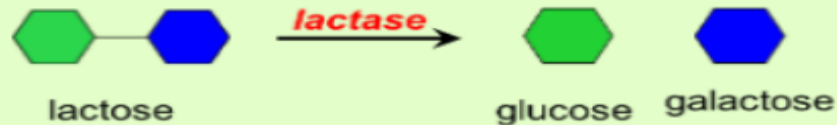
- The rest of the work of carbohydrate digestion is done by enzymes produced by the enterocytes, the cells lining the small intestine.
- The inner wall of the small intestine is actually composed of large circular folds, lined with many villi, the surface of which are made up of microvilli.
- All of this gives the small intestine a huge surface area for absorption.)

- By the end of this process of enzymatic digestion, we're left with three monosaccharides: glucose, fructose, and galactose. These can now be absorbed across the enterocytes of the small intestine and into the bloodstream to be transported to the liver.

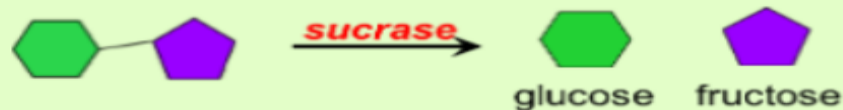
- Maltose is digested by **maltase**, forming 2 glucose molecules

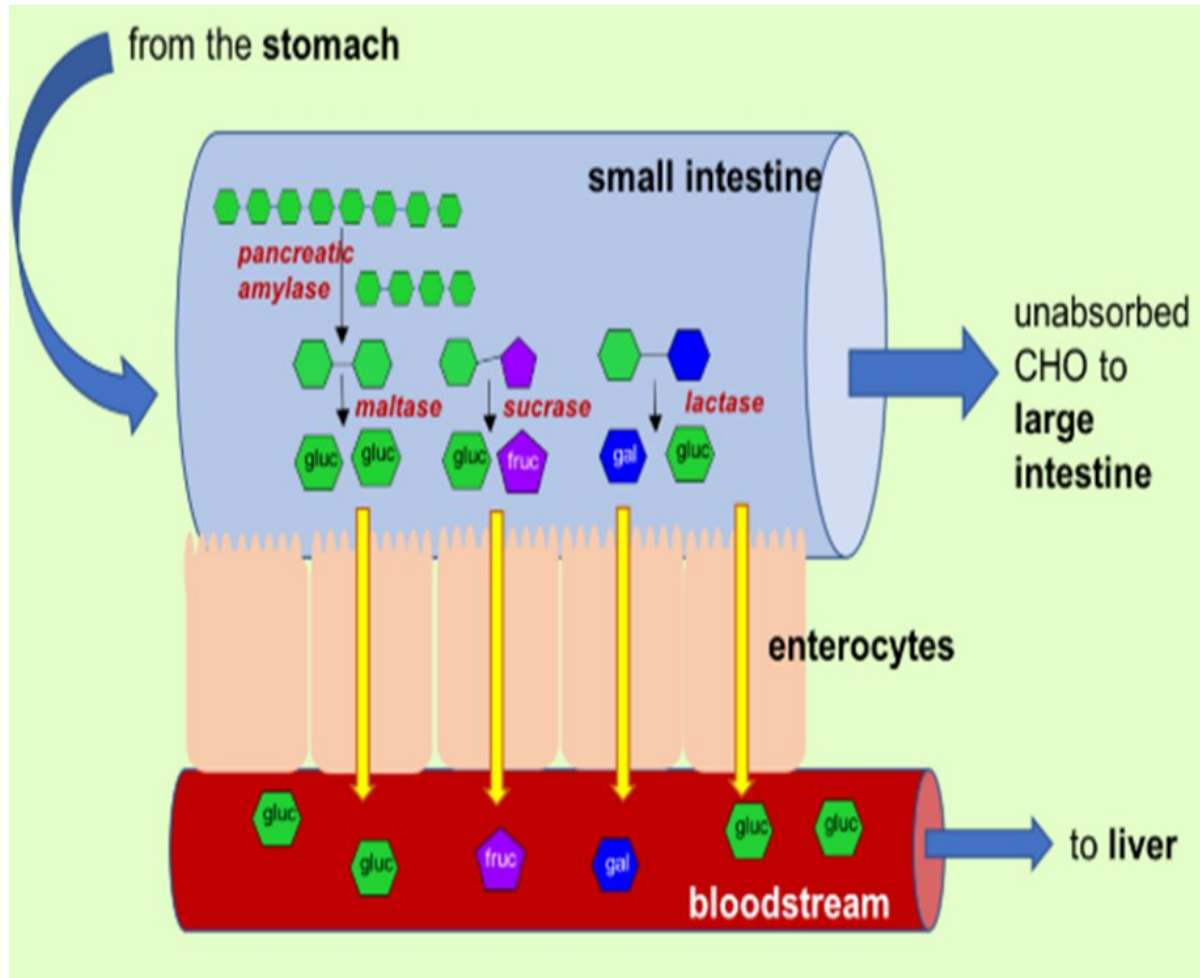


- Lactose is digested by **lactase**, forming glucose and galactose



- Sucrose is digested by **sucrase**, forming glucose and fructose





- Fructose and galactose are converted to glucose in the liver. Once absorbed carbohydrates pass through the liver, glucose is the main form of carbohydrate circulating in the bloodstream.

- Important functions of carbohydrates
 - Carbohydrates in the body function primarily in the form of glucose

[1] Source of energy:

- Glucose is a major source of energy for all the body cells.
- One gram of carbohydrate provides 4Kcal.
- RBCs are particularly dependent on glucose.
- It is also indispensable for the maintenance of functional integrity of the nerve tissue and under normal circumstances, it is the sole source of energy for the brain.
- Similarly, glucose is important for heart muscles. Although fatty acids are the preferred regular fuel of heart muscle, glycogen in cardiac muscle is an important emergency source of contractile energy.
- In a damaged heart, poor glycogen stores or low carbohydrate intake may cause cardiac symptoms of angina.

[2] Protein sparing effect:

- Carbohydrates help in regulating the protein metabolism.
- Presence of sufficient carbohydrates helps maintain lean muscle mass, especially during periods of caloric restriction, weight loss, or in clinical settings to prevent malnutrition.
- This protein sparing action allows the major portions of protein to be used for its basic structural purpose of tissue building.
- Therefore, patients who are unable to eat are temporarily administered 5% glucose solution intravenously.

[3] Antiketogenic effect:

- Presence of carbohydrates is necessary for normal fat metabolism.
- In the absence of sufficient carbohydrates, larger amounts of fat are used for energy than the body is equipped to handle.
- This results in incomplete oxidation and accumulation of ketone bodies. This may in turn lead to acidosis, sodium imbalance and dehydration.
- In extreme conditions such as starvation (carbohydrates are inadequate) and uncontrolled diabetes (where carbohydrates are unavailable for energy needs), ketoacidosis is a common complication.

[4] Excretion of toxins

- Glucuronic acid, a metabolite of glucose, combines with chemical and bacterial toxins and some normal metabolites in the liver and thereby helps in their excretion.

[5] Act as precursors

- Carbohydrates and their derivatives serve as precursors to compounds such as nucleic acid, connective tissue matrix and galactoside of nerve tissue.

[6] Overall positive health:

- Non-glycemic carbohydrates including non-starch polysaccharides are beneficial for the function and physiology of gastrointestinal tract and thus have a positive effect on the overall health.

Functions of Carbohydrates

- Supplies energy
- Protein sparing function
- Oxidation of fats
- Regulating blood glucose
- Facilitates bowel movements

